Zhang Xinlei

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EDUCATION

Nanyang Technological University, Singapore Aug 2025 – Jun 2026

Master of Science in Biomedical Engineering

Selected coursework: In Vitro Diagnostics and Bio-nanotechnology; Polymeric Materials for Biomedical Applications; Drug Delivery and Nanomedicine; Tissue Engineering and Regenerative Medicine.

Suzhou University of Science and Technology, China Sep 2021 – Jun 2025

Bachelor of Engineering in Environmental Science

RESEARCH EXPERIENCE

Research Assistant | Nanyang Technological University, Singapore

Sep 2026 – Present

Lectin-Mediated Magnetic Bacterial Enrichment Toward Rapid Antimicrobial Susceptibility Testing

- Investigating Fc-MBL-functionalized magnetic beads for the capture and enrichment of Escherichia coli and Staphylococcus aureus at low bacterial concentrations.
- Assessing bacterial recovery and capture efficiency through magnetic separation and culture-based colony-forming unit (CFU) enumeration.
- Comparing capture performance between Gram-positive and Gram-negative bacterial models to investigate species-dependent differences and guide optimization of the bacterial enrichment workflow.

Master's Research Project | Nanyang Technological University, Singapore

Sep 2025 – Aug 2026

Functional Gelatin–Chitosan Films for Food Preservation

- Prepared biodegradable gelatin–chitosan composite films and investigated tannic acid–zinc modifications for functional packaging applications.
- Evaluated wettability, FTIR spectra, water-vapor barrier properties, antioxidant activity (DPPH/ABTS), and antibacterial performance.
- Compared formulations through fresh-cut apple preservation experiments, including weight loss and browning assessments.

Undergraduate Research | Suzhou University of Science and Technology

Mar 2023 – Jun 2025

Isolation and Functional Characterization of Plant Growth-Promoting Rhizobacteria (PGPR)

- Isolated and screened plant growth-promoting rhizobacteria associated with rice lines tolerant to low nitrogen and phosphorus, using culture-based techniques and 16S rRNA-based identification.
- Investigated *Bacillus altitudinis* Hxx04 and its plant-growth-promoting potential; the study reported enhanced rice seedling growth and nitrogen/phosphorus uptake in pot experiments.

Undergraduate Research | Suzhou University of Science and Technology

Sep 2023 – Jun 2025

- *Zebrafish Neurotoxicity under Micro-/Nanoplastic and Silver Nanoparticle Co-exposure*

Conducted zebrafish exposure experiments starting at 6 hours post-fertilization (hpf) for a study of combined polystyrene micro-/nanosphere and silver nanoparticle toxicity.

• Performed larval locomotion assays using EthoVision XT (distance and velocity tracking) and generated behavioral data for downstream team analysis.

Undergraduate Research | Suzhou University of Science and Technology, China

Sep 2024 – Jun 2025

Isolation and Characterization of Sethoxydim-Degrading Bacteria

- Isolated and characterized four bacterial strains from contaminated soil using enrichment culture, biochemical characterization, and 16S rRNA gene sequencing.
- Identified *Delftia sp.* Z4 as a promising degrader, with 65.6% sethoxydim degradation quantified by HPLC.

PUBLICATIONS

Xinlei Zhang, Yuxin Fan, Xintong Jiang, Yuxi Yang, Hongyan Shen, Shan Zhuang, Jianheng Wang, Xinyu Liu, Jing Bai, Yanjie Qi, Dongfei Han. Isolation, identification, and application of plant growth-promoting rhizobacteria from backcross inbred rice populations with tolerance to low nitrogen and low phosphorus. *Acta Microbiologica Sinica*. 2026, 66(4): 1819–1838. DOI: 10.13343/j.cnki.wsxb.20250633

Jie Song, Qian Pu, Chen Chen, Xingcheng Liu, **Xinlei Zhang**, Zejun Wang, Jin Yan, Xuedong Wang, Huili Wang, Qiuhui Qian. Neurological Outcomes of Joint Exposure to Polystyrene Micro/Nanospheres and Silver Nanoparticles in Zebrafish. *Environmental Health Perspectives*. 2025, 133(5): 057007. DOI: 10.1289/EHP14873

Research contribution: Conducted zebrafish exposure experiments and larval locomotion assays using EthoVision XT.

Xinlei Zhang, Wenru Liu. Progress on nitrogen removal driving mechanism and influencing factors of Feammox. *Industrial Water Treatment*. 2025, 45(5): 1–11. DOI: 10.19965/j.cnki.iwt.2024-0526

PATENT

[First inventor] Formaldehyde Catalytic Manganese-Oxygen Composite Nano Filter Material and Its Preparation Method and Application. CNIPA patent application, application no. 2024104079088.

TECHNICAL SKILLS

Microbiology & bacterial assays: Bacterial culture and isolation; OD_{600} measurements; serial dilution; CFU enumeration; magnetic bead-based capture; 16S rRNA-based identification.

Biomaterials & characterization: Gelatin–chitosan film preparation; tannic acid–zinc modification; antibacterial and antioxidant (DPPH/ABTS) assays; contact-angle, FTIR, and water-vapor barrier analysis.

Analytical & data tools: HPLC-based quantification; SEM, XRD, XPS and BET characterization experience; GraphPad Prism, Origin, R/RStudio, Python/Jupyter Notebook, MATLAB.

Languages: Mandarin (native); English.

HONORS & AWARDS

- University First-Class Scholarship (rank 1/81), 2023–2024 academic year
- University Third-Class Scholarship (rank 19/81), 2022–2023 academic year
- National-Level College Student Innovation Training Project (2025)
- University-Level Outstanding Project Award, National College Student Energy Conservation and Emission Reduction Competition (2024)

REFERENCES

Prof. Hongwei Duan | hduan@ntu.edu.sg | MSc research supervisor
Nanyang Technological University

Prof. Dongfei Han | dongfeihan@usts.edu.cn | Undergraduate supervisor
Suzhou University of Science and Technology